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Interface characteristic of Zr-based metallic glass and copper by laser pulse welding

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ABSTRACT

The crystallization behavior of amorphous alloys during welding is more complex than that of traditional metals. In this paper, the Zr-based amorphous alloy and copper crystals were welded by pulsed laser butt welding method, the crystallization behavior of the amorphous alloy/crystalline metal welding interface is deeply studied. The results show that Zr_2Cu and Zr_2Ni crystal phases appear at the interface between Zr based amorphous alloy and copper, and a Cu rich region is formed at the amorphous side of the interface. A small amount of Zr, Ni, Ti diffuses to the Cu side, forming ZrO_2, TiO_2, CuO and other oxides on the surface of the sample. These oxides will cause changes in the voltage potential at the interface.

1. Introduction

Bulk metallic glasses (BMGs) show excellent mechanical, physical and chemical properties compared to traditional metals due to their short-range order and long-range disorder in microstructure, such as high strength, high elasticity, excellent soft magnetic properties and good corrosion resistance [1,2], it has been widely used in aerospace, chemical, military industry and other fields [3,4]. So far, in the developed BMGs of magnesium-based, lanthanum-based, titanium-based, zirconium-based and other systems [5–8], the high strength and good superplasticity of Zr-based amorphous alloys have attracted extensive attention [9]. It is simple to prepare amorphous alloys with excellent properties because zirconium-based amorphous alloys have a strong glass-forming ability and a wide supercooled liquid phase region, with a critical cooling rate as low as 1 K/s [10] and a supercooled liquid phase region that is extremely wide over 100 K [11]. Bulk amorphous alloy preparation is still a highly challenging and expensive process. Therefore, in order to maximize the uniqueness of amorphous alloys, researchers prepare composites by alloying or introducing crystalline phases.

At present, laser processing is one of the most promising processing methods to connect BMG to crystal alloys. During the laser machining process, the cooling rate of the molten pool can reach 10^5~10^8 K/s [12], which is much higher than the critical cooling rate of most amorphous alloys. Therefore, it is feasible to produce amorphous alloys. Researchers prepared amorphous/crystalline composite layers with excellent performance by laser remelting [13]. More researchers are concentrating on laser welding, laser cladding, and the development of additive manufacturing for bulk amorphous alloys as a result of the versatility of laser processing and the excellent performance of amorphous alloys in bulk [14]. Zhang et al. successfully prepared Mg-based amorphous layers with standard composition $Mg_{65}Zn_{30}Ca_5$ using laser remelting method [15], the effect of laser scanning speed on the microstructure of Mg-Zn-Ca amorphous layer during laser remelting was systematically investigated. Studies have shown that an amorphous phase can be formed when the laser scanning speed is kept above 1000 mm/min. The remelting layer is mainly composed of gradient structure amorphous composite layer, transition layer and crystalline matrix.

In the welding process of amorphous alloys and crystalline materials, the cooling rate can not be well controlled during solidification, which leads to crystallization of amorphous alloys in the welding zone and heat affected zone. At the same time, the melting of amorphous alloys will lead to the formation of intermetallic compounds at the interface between amorphous and conventional metals. Crystallization at the joint and the formation of intermetallic compounds at the interface will make the amorphous alloy lose its original excellent properties. In order to effectively analyze the movement of atoms in the crystallization process of amorphous alloys and the diffusion behavior of atoms at the interface

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of welded joints, the researchers have carried out a computer virtual environment based on relevant theories, from various microscopic scales. Numerous simulations have been performed on the melting and solidification process, the formation energy of amorphous alloys, and the diffusion of atoms between welded joint interfaces [16–18]. It provides a powerful guide for in-depth understanding of the solid-liquid phase transition of amorphous alloys during solidification and crystallization and the diffusion between atoms during welding.

In this paper, Zr based amorphous alloy was connected to copper plate by laser welding, and Zr based amorphous and copper butt welding specimens were successfully prepared. The interface structure and chemical composition of Zr based amorphous and copper butt welding were analyzed by X-ray diffraction, scanning electron microscope and energy dispersive spectrometer, electron back scattering diffraction and atomic force microscopy. The mechanical properties of the welded joint were characterized by microhardness and nano indentation techniques.

2. Materials and experimental detail

2.1. Materials

The $Zr_{41.2}Ti_{13.8}Cu_{12.5}Ni_{10}Be_{22.5}$ BMG alloy ingot with namely of with the diameter of 10 mm was prepared by injection cast into copper mould in a non-consumable vacuum arc furnace suction. The BMG alloy ingot was cut by electric spark wire cutting machine into small disc with the thickness of 3 mm. Before welding, the XRD test was employed to detect whether the amorphous material is completely amorphous structure. A clear widen peak and no other strong diffraction peaks were visible in the XRD pattern, which was indicative of an amorphous substance. It was indicated that $Zr_{41.2}Ti_{13.8}Cu_{12.5}Ni_{10}Be_{22.5}$ alloy is amorphous alloy. The commercial C11000 copper plate with the thickness of 2 mm was used to copper base metal.

2.2. Experimental procedure

To obtain a high quality welding sample, the base materials were cleaned using a sandpaper with a particle size of 2000 to remove the oxides on the surface before welding and then cleaned with acetone. Above mentioned small disc of $Zr_{41.2}Ti_{13.8}Cu_{12.5}Ni_{10}Be_{22.5}$ BMG were cut along the diameter direction, which was used to join copper plate with the dimension of 10 mm $\times$5 mm $\times$2 mm by pulse laser welding. A butt welded joint of $Zr_{41.2}Ti_{13.8}Cu_{12.5}Ni_{10}Be_{22.5}$ BMG and copper was obtained, and the macroscopic image of the joint is shown in Fig. 1. The schematic diagram and actual welding process diagram of pulse laser welding are shown in Fig. 2. In the pulse laser welding of Cu and Zr-based BMG, adjusting the distance between the focusing lens and the working platform controlled the spot diameter of 4 mm. In the experiment, the basic power is 90 W, the peak power is 1300 W, the duty cycle is 70%, the welding speed is 2 m/min and pulse frequency is10 Hz.

After welding, the specimen for the microstructure observation was obtained by cutting the welded sample vertical to the welding direction. The specimens for material characterization were prepared using the

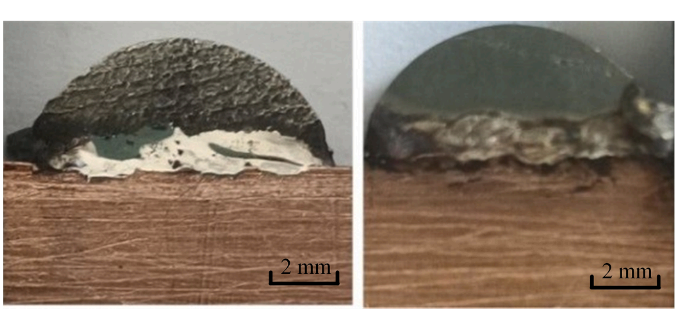

Fig. 1. Macroscopic image of welded joints.

standard metallographic technique. The different region structure of welded joint was characterized by X-ray diffractometer (XRD) using selected area micro diffraction with a Cu Kα radiation source. The working voltage and working current were 40 kV and 120 mA, respectively. Using a scanning speed of 8°/min to measure 2°~110° region and step of 0.02°. The microstructure examinations of the samples were carried out using optical microscopy (OM), scanning electron microscopy (SEM) and electron backscatter diffraction (EBSD). The samples polished by argon ion were characterized by EBSD for 100 μm×80 μm area in a 1 μm step. Then OIM analysis software was employed to process EBSD data. In addition, using X-ray photoelectron spectroscopy (XPS) to characterize compounds composition, and atomic force microscope (AFM) and SKPFM was adopted to obtain the three-dimensional morphology and corresponding volt potential of Cu/BMG interface.

The hardness distribution of welded joint was determined along the center line of the cross-section using Vicker microhardness tester with an applied load of 100 g and load time of 10 s. A Nanoindentation was conducted across the interface using Hysitron TI 950 with the maximum dentation depth of 50 nm at a loading rate of 1.6 mN/s and load time of 2 s. Three sets of parallel experiments were conducted under the same parameters for microhardness and nanoindentation.

3. Results

3.1. Microstructure of laser welded joint

The XRD pattern of BMG/Cu butt welded joint at different region are shown in Fig. 3. It can be seen from XRD pattern that Zr-based BMG maintain amorphous structure, and the XRD pattern of Cu base metal presents the typical diffraction peaks of Cu. There are other crystal phase diffraction peaks except the Cu diffraction peak in the BMG/Cu interface, which are Zr_2Cu phase, Zr_2Ni phase. It is indicated that the BMG and copper react to form intermetallic compounds during the pulse laser welding.

Low magnification SEM image of interface between copper and $Zr_{41.2}Ti_{13.8}Cu_{12.5}Ni_{10}Be_{22.5}$ BMG as shown in Fig. 4(a). And no defects are detected in the joint indicating that the sound joining was achieved between the BMG and pure Cu by pulse laser welding. There are three region in the BMG/Cu interface, dark of Cu, bright of BMG and diffusion layer of mix area, respectively. The diffusion layer forms on the side of BMG near the top of the welded zone, which thickness is above 10 μm, while on the bottom of welded joint appears an obvious line profile of BMG/Cu.

For further study microstructure characterization of the diffusion layer, line scan analysis across the welded interface from Zr-based BMG side to Cu side was conducted. Fig. 4(b) shows line profile across the weld joint presenting concentrations of Zr, Ni, Cu and Ti. It is observed that obvious variation in the concentration distribution of elements in the weld metal except unobvious diffusion of Ni in weld zone. Through the element line scanning distribution results, it can be seen that the rapid increase of Cu element appears in the feather like diffusion layer near the amorphous side, and the Cu element content of the diffusion layer gradually decreases when it gradually approaches the metal Cu. The change of Zr content is just opposite to that of Cu, and the content of Ti also fluctuates.

To further analyze the microstructure and chemical composition of BMG/Cu interface, the EDS point test are characterized the chemical composition of BMG/Cu interface. Fig. 4(c) displays the distribution of Zr, Ni, Cu and Ti elements at the BMG/Cu interface. According to Fig. 4 (a), it can be found that the edge of Cu/amorphous alloy interface is not smooth in fact, and some large protrusions can be observed. Cu fragments in a form similar to feather unevenly distributed in the amorphous alloy matrix. EDS map result shows a great amount of Cu element distributed in the diffusion layer, while there is less distribution of Zr element. Similarly, there are less distribution of Ti and Ni element.

The EDS point element analysis were conducted the content of Zr, Ni,

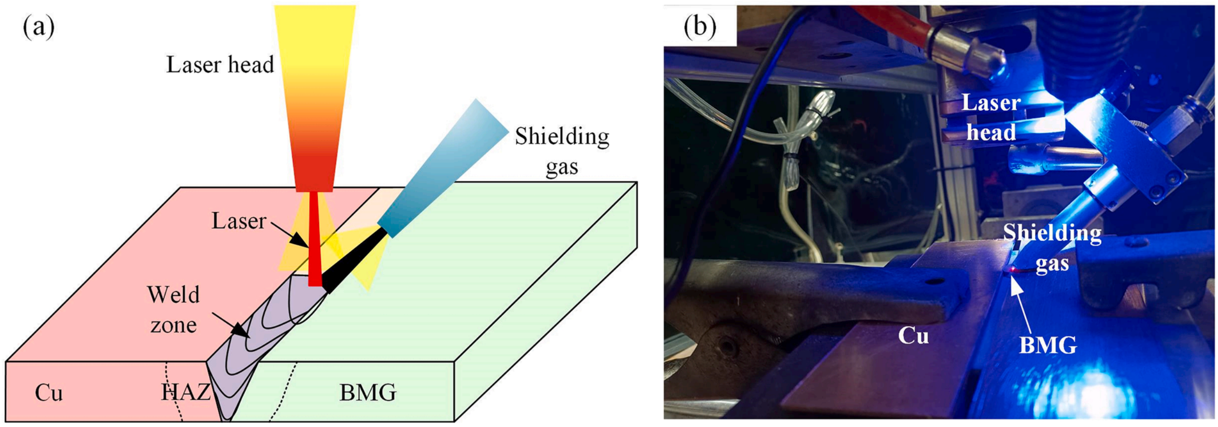

Fig. 2. (a) Schematic diagram of pulse laser welding and (b) practical welding process.

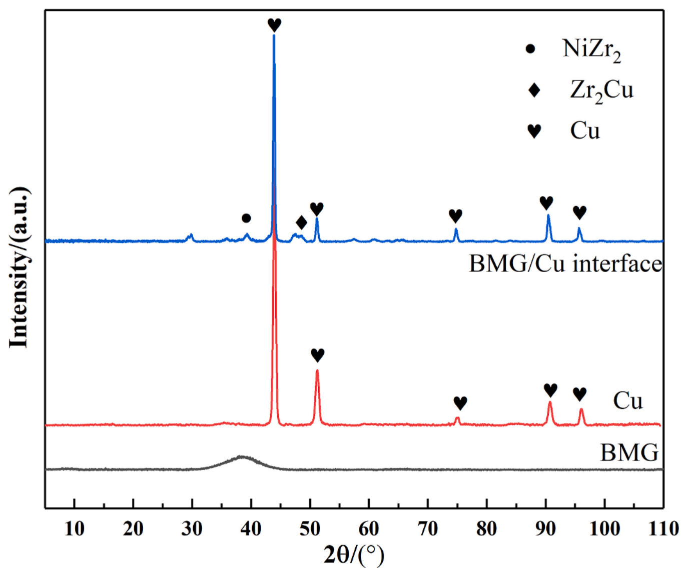

Fig. 3. XRD pattern of welded joint at different region.

Cu and Ti elements at the different positions of the joint marked with 1, 2, 3, 4 and 5, and the EDS result at these positions are listed in Table 1. The point 1 at the diffusion layer near the amorphous alloy, which consists of 69.5% Cu, 17.7% Zr, 9.0% Ti and 3.8% Ni. The point 2 at the amorphous consists of 49.1% Cu, 34.0% Zr, 10.5% Ti, 6.4% Ni. When the point at the diffusion layer near the copper side, the content of Cu rapidly increases and closes to the content of Cu base metal. Remarkably, the content of Cu element in the amorphous alloy after welding obviously higher than as-cast amorphous alloy (12.5% Cu). It is illustrated that diffusion of Cu element into amorphous alloy not only forms diffusion layer near the BMG/Cu interface, and also increases Cu content of amorphous alloy.

The result of SEM and EDS characterization in the BMG/Cu interface illustrated that the formation of a metallurgical bonding interface between Zr-based BMG and copper by pulse laser welding. A great number of Cu element diffuse into Zr-based BMG and form copper rich area in the Zr-based BMG. This is because the laser beam concentrates on the Zr-based BMG during pulse laser welding, leading to the Zr-based BMG is positioned in the supercooled liquid region and the equilibrium viscosity of the alloy decreases sharply [19,20]. The decrease of total free energy for BMG, which provides diffusion path to Cu atoms. In additional, due to the heat of laser beam does not reach the melting temperature of metal copper, it is difficult for amorphous alloy to diffuse to the side of metal Cu. Moreover, since the atomic diffusion rate from metal Cu to amorphous alloy is much faster than the opposite process, the diffusion layer located on the side of amorphous alloy.

The XPS test was conducted at the BMG/Cu interface, which proved the presence of C, Zr, Ti, Ni and Cu element in the BMG/Cu interface. Fig. 5 shows the XPS survey spectrum and the individual high resolution spectrum lines of all constituents of BMG/Cu interface. There are Zr 3d, Ti 2p, Cu 2p, O 1s, C 1s (which is induced by the carbon contamination) in the BMG/Cu interface, as shown in Fig. 5(a). XPS did not detected the Ni and Be element due to the content is small. In the high resolution spectrum lines of Zr 3d, there is a 2.4 eV spin orbit splitting between Zr $3d_{5/2}$ line and Zr $3d_{3/2}$ line. The double peak of Zr^{4+} are identified at the binding energy of 183.1 eV and 185.5 eV, respectively, which corresponds ZrO_2 oxide [21,22], as shown in Fig. 5(b). For the Ti 2p, Ti $2p_{2/3}$ and Ti $2p_{1/2}$ of Ti^{4+} are distinguished at the binding energy of 459.4 eV and 465 eV, respectively, which corresponds TiO_2 oxide [23], as shown in Fig. 5(c). Moreover, Ti^{3+} are also distinguished at the binding energy of 469 eV, which corresponds Ti_2O_3 oxide [24]. Fig. 5(d) shows the high resolution spectrum line of Cu 2p, the double peak of Cu^{2+} are identified at the binging energy of 931.7 eV and 952.2 eV, respectively, corresponding CuO oxide [25].

From the XPS high-resolution spectrum, it can be seen that Zr 3d, Ti 2p, Ni 2p and Cu 2p are a combination of metal state and oxidation state, indicating that ZrO_2, TiO_2, CuO and NiO oxides exist on the surface of the BMG/Cu interface, which may be due to the higher oxygen content on surface. It is obviously found that the percentage content of Cu 2p, Zr 3d and Ti 2p is high, indicating that the interface contains high content of Zr, Ti and Cu elements, and the content of Ni element is very small. The oxides of ZrO_2, TiO_2 and CuO exist at the BMG/Cu interface. From the perspective of comprehensive strength, the natural oxides of Cu^{2+}, Zr^{4+} and Ti^{4+} are mainly formed on the surface of BMG/Cu interface. The Zr^{4+} and Ti^{4+} are more likely to form oxides in air due to their oxide have lower negative enthalpy of formation [26]. Related research shows that the active Zr migrates to the oxide film in the Cu-Zr-Al bulk metallic glass, and the less active copper diffuses to the oxide/metal interface [27]. In addition, the high content of Cu element also forms oxides at the interface.

The EBSD result of BMG/Cu interface and copper base metal are shown in Fig. 6. Fig. 6(a) shows the EBSD map in the Zr-based BMG/Cu interface. The black zone on right side of interface is the Zr-based BMG. Due to the disordered atomic structure of amorphous alloys, Kikuchi patterns cannot be obtained from amorphous alloys. In additional, a great number of fine grains of copper appear in the BMG. The formation of fine Cu grains is caused by the Cu grain structure had not enough growth space. The result shows that the Zr-based BMG still maintained amorphous structure after pulse laser welding, and the content of crystal phase is less than 0.1%. The Cu atoms diffuse into the Zr-based BMG, which realizes the connection of Zr-based BMG and copper. The left side

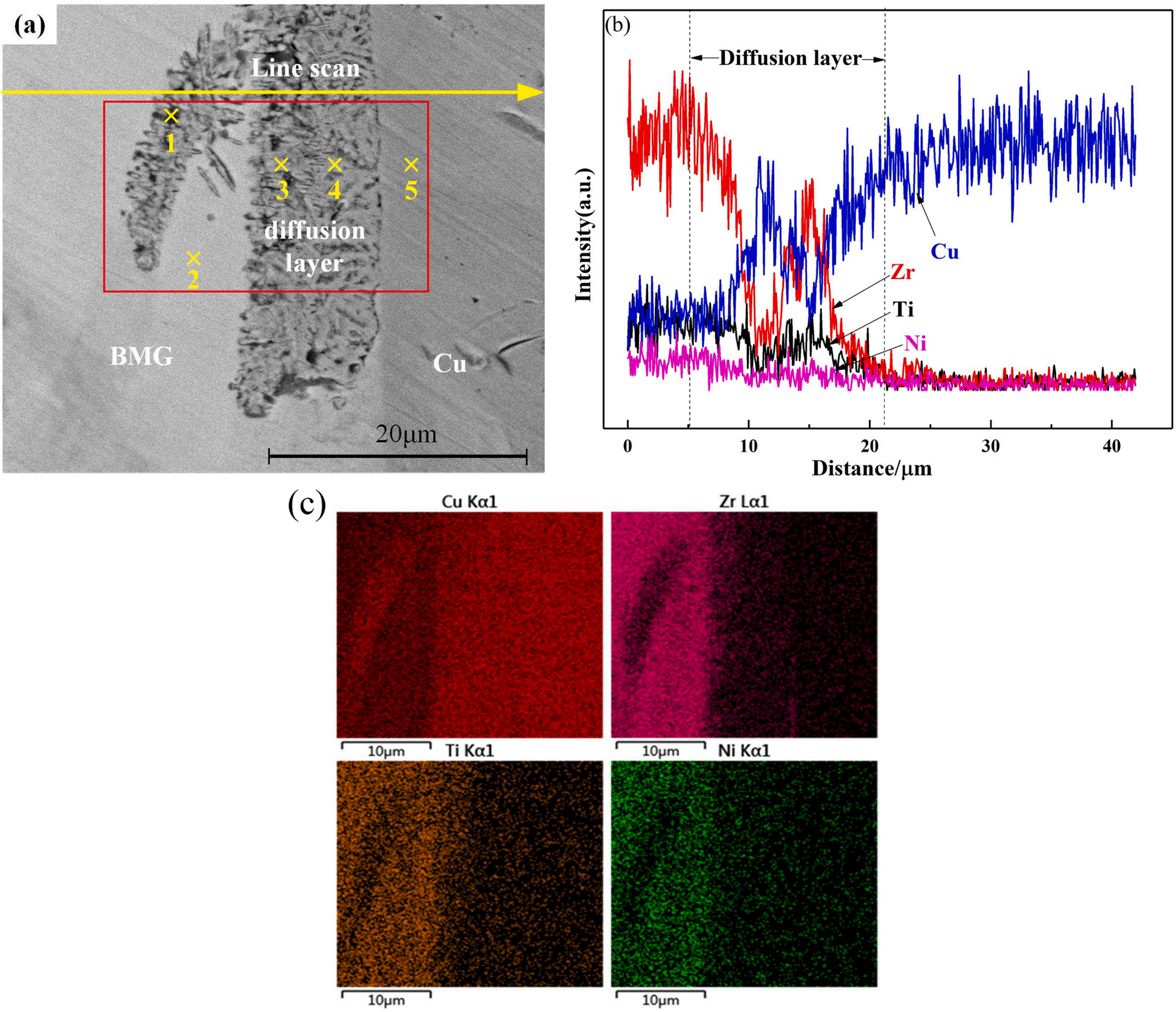

Fig. 4. (a) Backscattered electrons micrograph, (b) EDS analysis across the joint interface and (c) EDS point scan result.

Table 1
The EDS point element analysis result at different region (at.%).

EDS spot	Zr	Cu	Ti	Ni
1	17.7	69.5	9.0	3.8
2	34.0	49.1	10.5	6.4
3	4.8	90.8	3.1	1.3
4	1.8	97.1	0.8	0.2
5	0.2	99.8	0.0	0.0

of Zr-based BMG/Cu interface is the EBSD map of Cu. It can be found that the Cu side appears abnormally grown grains. The grains unevenly distribute in Cu matrix. Compared with the Cu grain size on the Zr-based BMG/Cu interface and Cu matrix, Cu grain in the Zr-based BMG/Cu interface has bigger size. It is generally believed that the uneven grain on the copper side is caused by the incomplete recrystallization during the heating process [28]. During the EBSD analysis, the grain size distribution of Zr-based BMG/Cu interface was measured, as shown in Fig. 6(b). The Cu grain in the diffusion layer is very small, dueing to the growth of Cu gain is limited by the amorphous alloy. In the Zr-based BMG/Cu interface, a great number of grains are located within 1 μm and the percentage is larger than 90%. Additionally, the grain size larger than 1 μm located at Cu side.

3.2. Analysis of interface morphology

To further analyze the interface behavior of Zr-based BMG and Cu. AFM characterized the three-dimensional morphology of BMG/Cu interface and its corresponding Volt potential reflection result. Fig. 7(a) shows the three-dimension morphology of BMG/Cu interface without diffusion layer. The height of copper, BMG/Cu interface and Zr-based BMG are different. Obviously, the height of BMG/Cu interface is higher than base metal on both sides and the interface surface is nonuniform. Fig. 7(b) shows the volt potential at the white line in Fig. 7(a). It can be seen that the volt potential at the interface is higher than the base metal on both sides, and the maximum volt potential at the interface is about 884 mV. The three-dimensional morphology of BMG/Cu interface including diffusion layer is shown in Fig. 7(c). The interface composes of a large number of uneven particles. Its surface roughness is higher than BMG/Cu interface without diffusion layer, which is caused

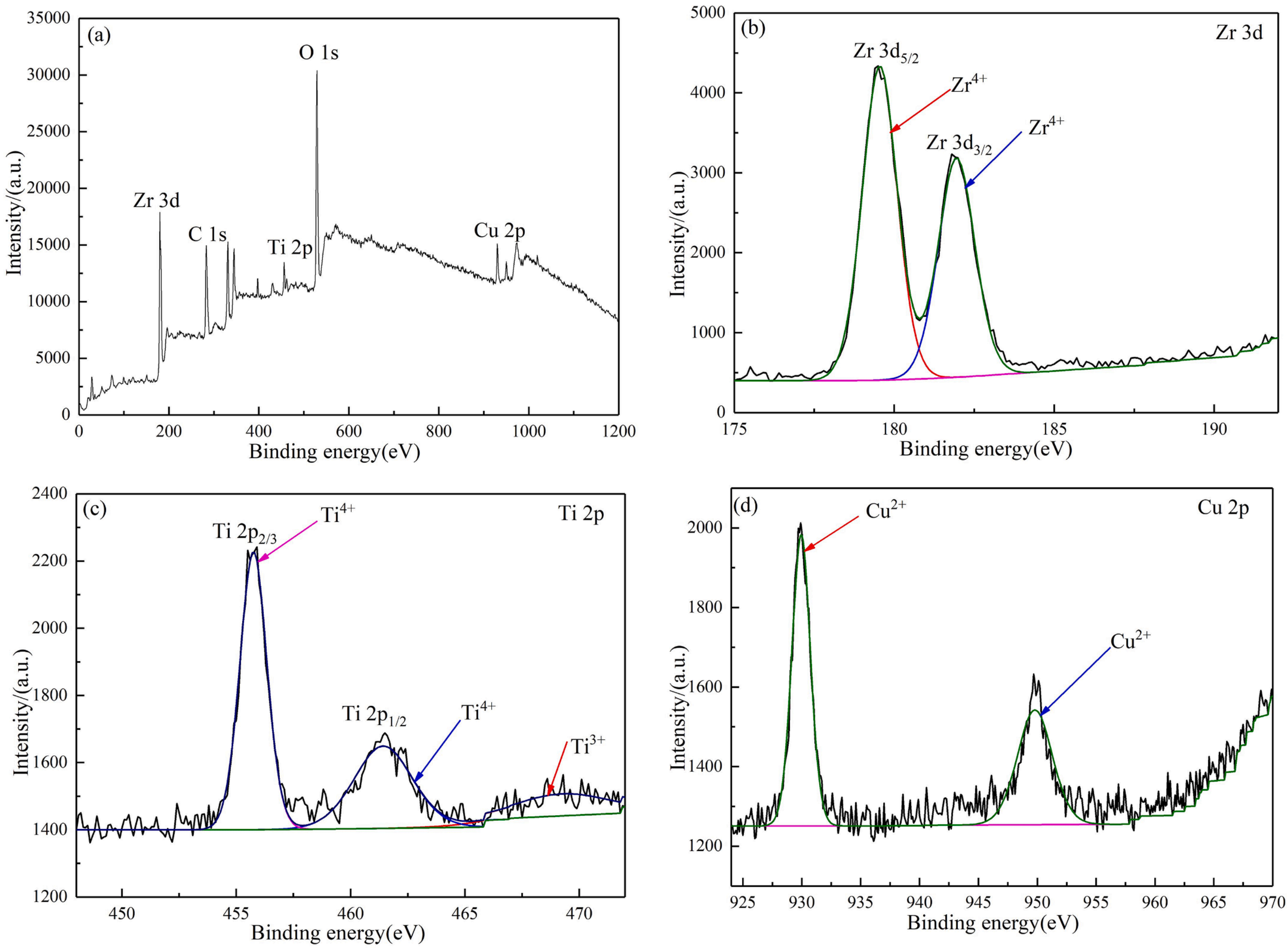

Fig. 5. XPS spectra of BMG/Cu interface: (a) XPS survey spectrum and (b) Zr 3d, (c) Ti 2p, (d) Cu 2p.

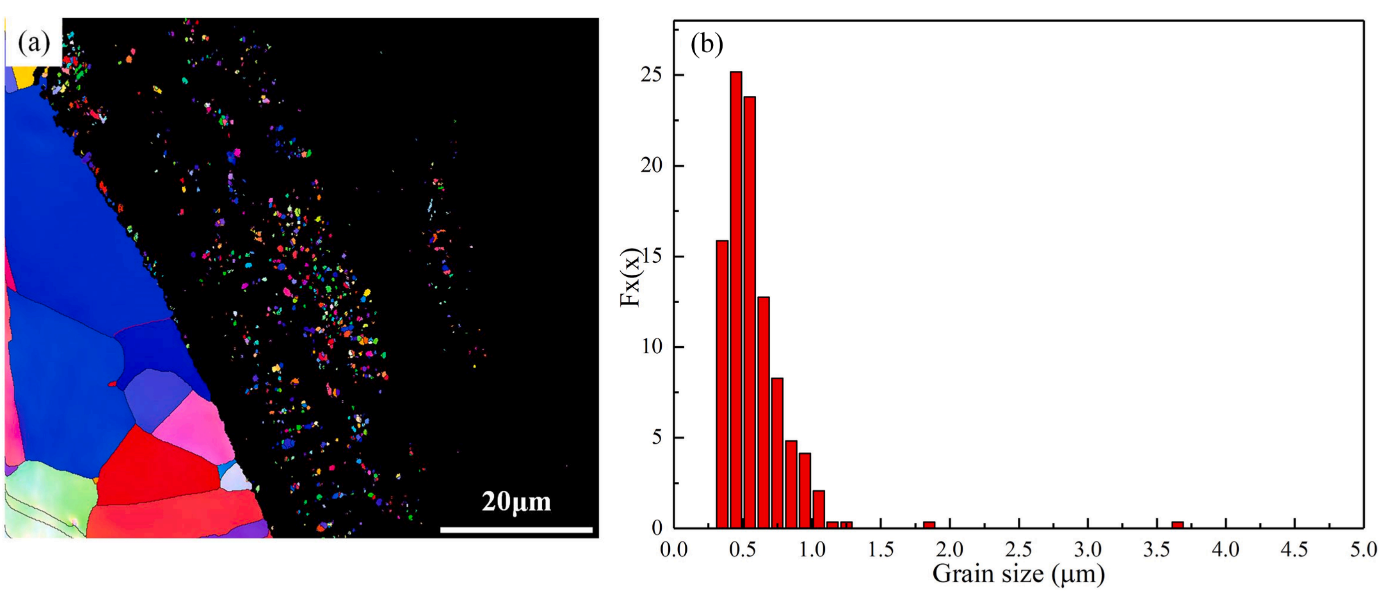

Fig. 6. EBSD result of welded joint: (a) BMG/Cu interface and (b) the grain size distribution in BMG/Cu interface.

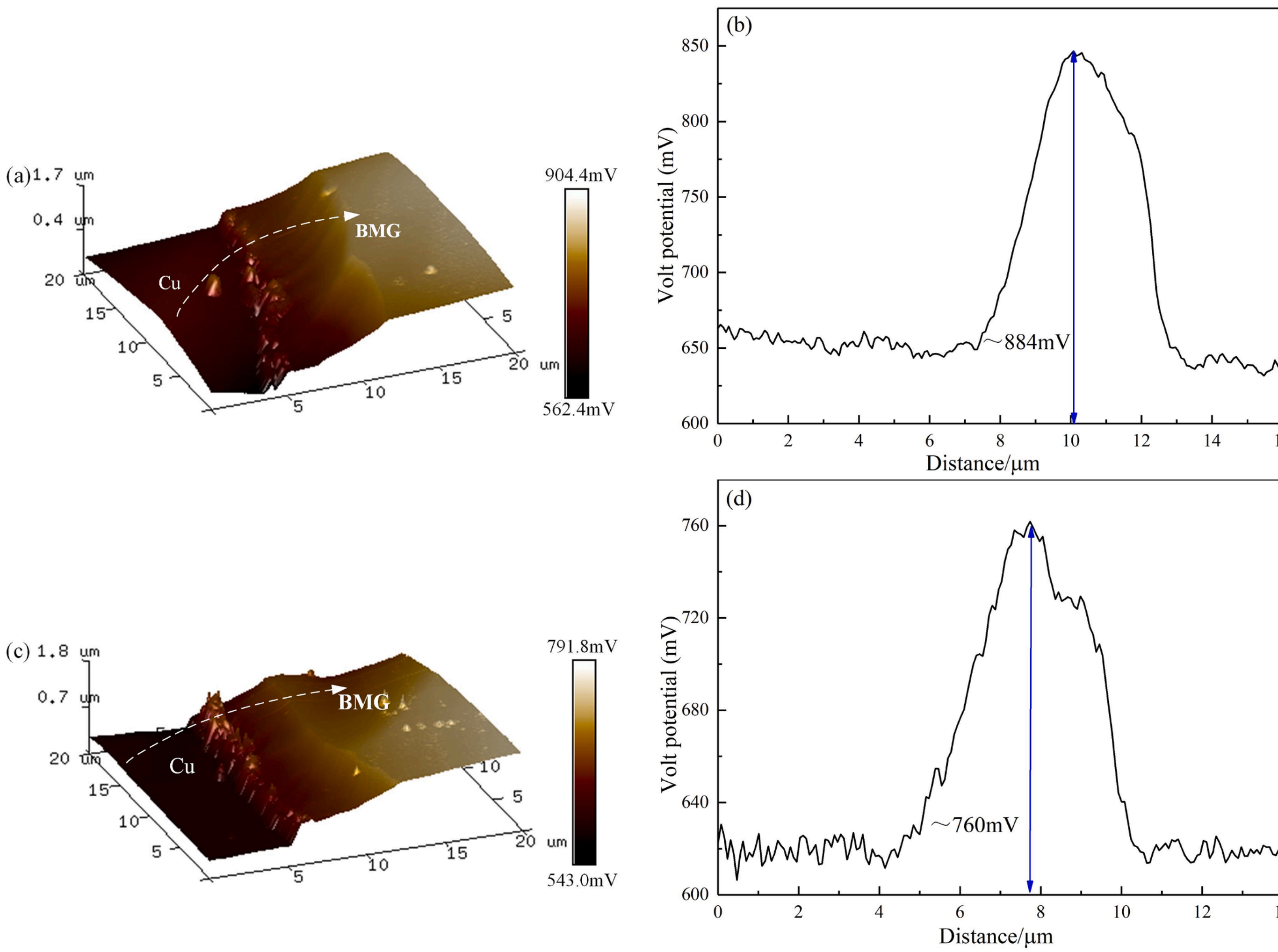

Fig. 7. AFM analysis result: (a) three-dimensional morphology of BMG/Cu interface, (b) Corresponding volt potential at white line, (c) three-dimensional morphology of diffusion layer and (d) Corresponding volt potential at white line.

by the Cu atoms diffusion into Zr-based BMG forming nanoparticles. Corresponding volt potential mapping results of the white line in Fig. 7(c) is shown in Fig. 7(d), the volt potential at the interface is still higher than the base metal on both sides, and the maximum volt potential at the interface is about 760 mV.

The AFM results show that the volt potential at the interface is higher than the base metal on both sides, which is caused by the presence of oxide at the interface. As reported by Wang et al. [29], the volt potential

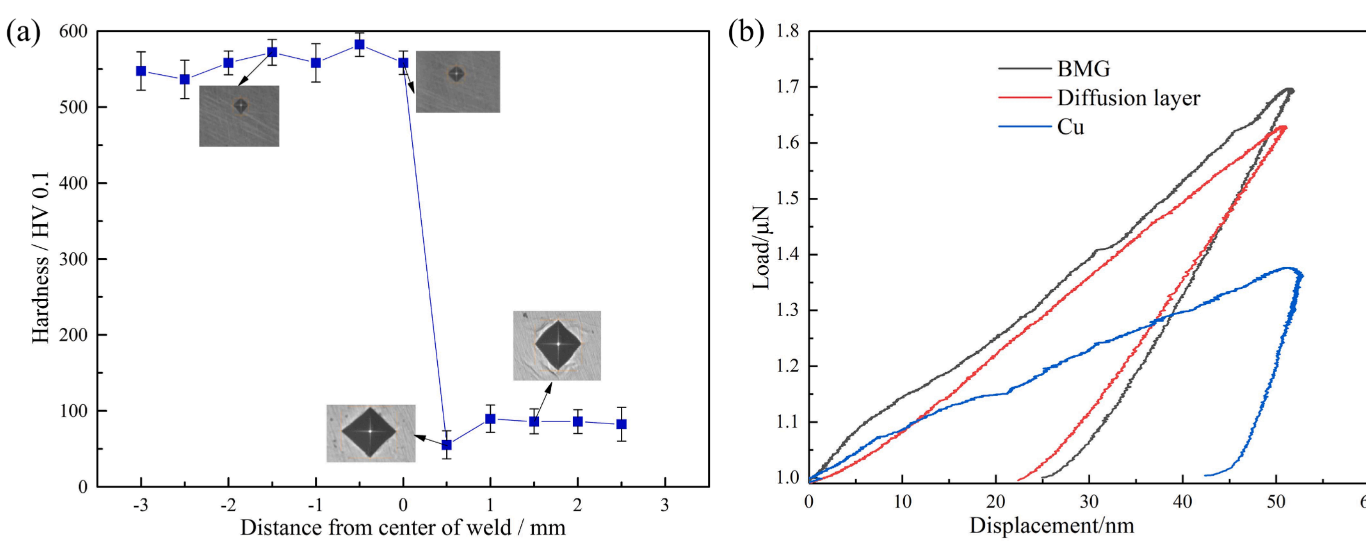

Fig. 8. (a) The hardness distribution of welded joint and (b) nanohardness load-depth curves between copper plate, diffusion layer and BMG.

measured by SKPFM is mainly affected by the type and content of elements, containing more precious metal elements or easy to form oxide layer on the surface have higher volt potential. As can been known from the result of XPS (Fig. 5), the BMG/Cu interface has more oxide, which made interface has higher volt potential. Moreover, the volt potential of BMG/Cu interface is higher than diffusion layer. The reason is that the diffusion layer had more Cu atoms, Cu atom can decrease volt potential [30].

3.3. Mechanical property

Fig. 8(a) displays the hardness distribution of butt welded joint for Zr-based BMG and copper. According to the variation of hardness, the hardness distribution could be divided three areas of amorphous alloy, diffusion layer and copper base metal, respectively. The hardness of amorphous alloy side reaches to 550 HV, which reflects the high strength and high hardness of amorphous alloy. The hardness of amorphous alloy increase near the BMG/Cu interface and the highest hardness reach to 600 HV. The hardness of copper near the BMG/Cu interface decrease to about 45 HV. In other region of copper base metal, its hardness is about 80 HV. Remarkably, near the BMG/Cu interface, the hardness of amorphous alloy is higher than Zr-based BMG base metal and the hardness of copper is lower than copper base metal. According to the result of EBSD, Cu atoms diffuse into the amorphous alloy, which plays the role of dispersion strengthening and increases the hardness of the amorphous alloy near the interface. The grains of copper near the BMG/Cu interface appear abnormally grown, which leads to the hardness decrease.

For further analyze the hardness near the BMG/Cu interface, nanoindentation tests were carried out in different areas near the interface. Fig. 8(b) shows the load penetration depth curve with indentation depth of 50 nm in different areas of the welded joint. The unsmooth load-displacement curve may be caused by too small strain rates. Wang et al.'s research results on nanoindentation of $Zr_{64.13}Cu_{15.75}Ni_{10.12}Al_{10}$ amorphous alloy can explain this phenomenon [31]. The hardness and elastic modulus of the diffusion layer are 5.99 Gpa and 83.58 Gpa respectively, the hardness and elastic modulus of the amorphous alloy side are 5.87 Gpa and 107.54 Gpa respectively, and the hardness and elastic modulus of the copper side are 1.49 Gpa and 152.64 Gpa respectively. The penetration depth of the diffusion layer is significantly smaller than that of metallic copper, but greater than that of the amorphous alloy matrix. This means that the interface is not a weak connection and its deformation resistance is higher than that of metallic copper. In other words, the interface between the amorphous alloy matrix and the copper metal is well bonded during pulse laser welding. Similarly, due to the hardening effect of copper atoms diffusion into amorphous alloy, the hardness of diffusion layer is higher than that of copper and amorphous alloy. Due to the low modulus and high hardness of amorphous alloy, the hardness of amorphous alloy is much higher than that of copper plate, but the modulus of amorphous alloy is lower than that of copper plate.

4. Discussion

From the above results, it can be seen that crystalline phase appears at the interface of Zr-based amorphous alloy/copper connected by pulsed laser butt welding. The diffusion of amorphous atoms at the interface will generate oxides on the surface of the sample, and the existence of oxides will cause the change of interface voltage potential. The whole diffusion process can be illustrated in Fig. 9. In the first step, Cu is diffused into Zr-based BMG. In the second step, the diffused Cu atoms begin to crystallize. The third step is to form a metallurgical bonding layer on the amorphous side. Yu et al. used pulsed laser beam welding and brazing technology to weld $Zr_{44}Ti_{11}Ni_{10}Cu_{10}Be_{25}$ bulk metallic glass and copper [32]. The results showed that a diffusion layer was formed between the fusion zone and copper, and the diffusion of copper into BMG caused the change of Zr atom concentration. Wang et al. used laser impact welding to weld Fe-based amorphous alloy and copper [33], and no visible element diffusion and intermetallic phase were found in the welding interface, indicating that the atomic diffusion between the amorphous and metal welding interface is complex.

In order to explain the atomic diffusion phenomenon at the interface of Zr-based amorphous alloy/copper at the micro scale, a model of Zr-based amorphous alloy/copper interface was established using molecular dynamics method. According to the different diffusion degree at different positions from the heat affected zone to the weld center, the diffusion process at four temperatures of 600 K, 700 K, 800 K and 900 K was simulated. Fig. 10 (a) shows the concentration distribution of each atom in the interface model on the z-axis at 900 K. It can be seen that both crystalline and amorphous Cu atoms have obvious diffusion phenomenon, forming a concentration gradient, while amorphous Zr atoms have no obvious gradient change, indicating that the diffusion process is mainly the diffusion of crystalline copper atoms to the amorphous side, which basically conforms to the test results. Fig. 10 (b) shows the variation curve of diffusion layer thickness with temperature. It can be seen from the figure that the higher the temperature is, the greater the diffusion layer thickness and the more obvious the increase is. It shows that the closer to the center of the weld, the more obvious the diffusion phenomenon.

In the study of Zr-based amorphous alloy/copper interface, in addition to diffusion behavior, crystallization behavior is also the focus of the study. Therefore, Yu et al. also studied the structural transformation mechanism of the fusion zone (FS) and the heat affected zone (HAZ) through the C curve [32], and found that there was no intersection between the thermal cycle curve and the C curve of Tx during the solidification process, indicating that the material in the heat affected zone still maintained the amorphous structure during the pulsed laser welding process, which is consistent with the above experimental results. In

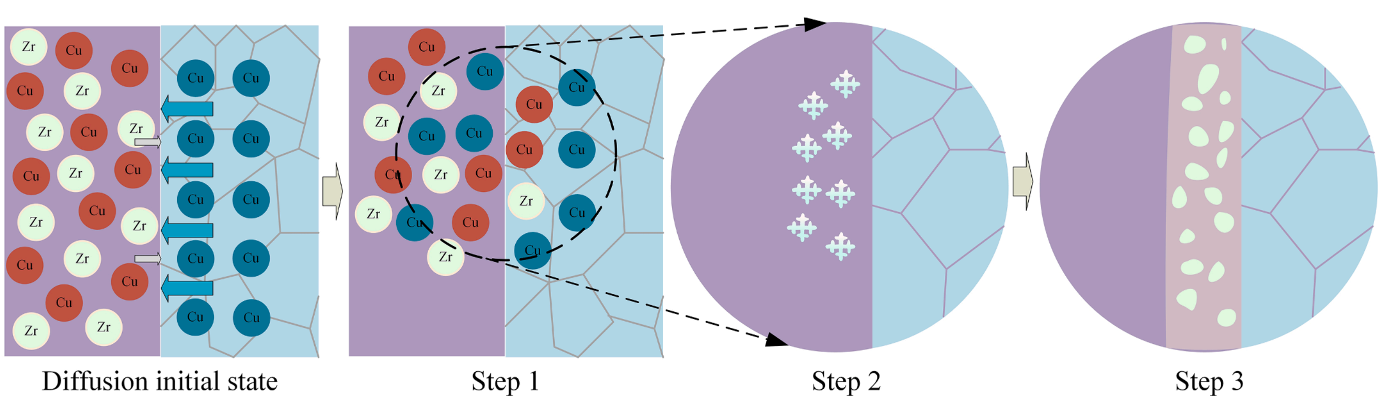

Fig. 9. Schematic diagram of atomic diffusion and crystallization process at amorphous/Cu interface

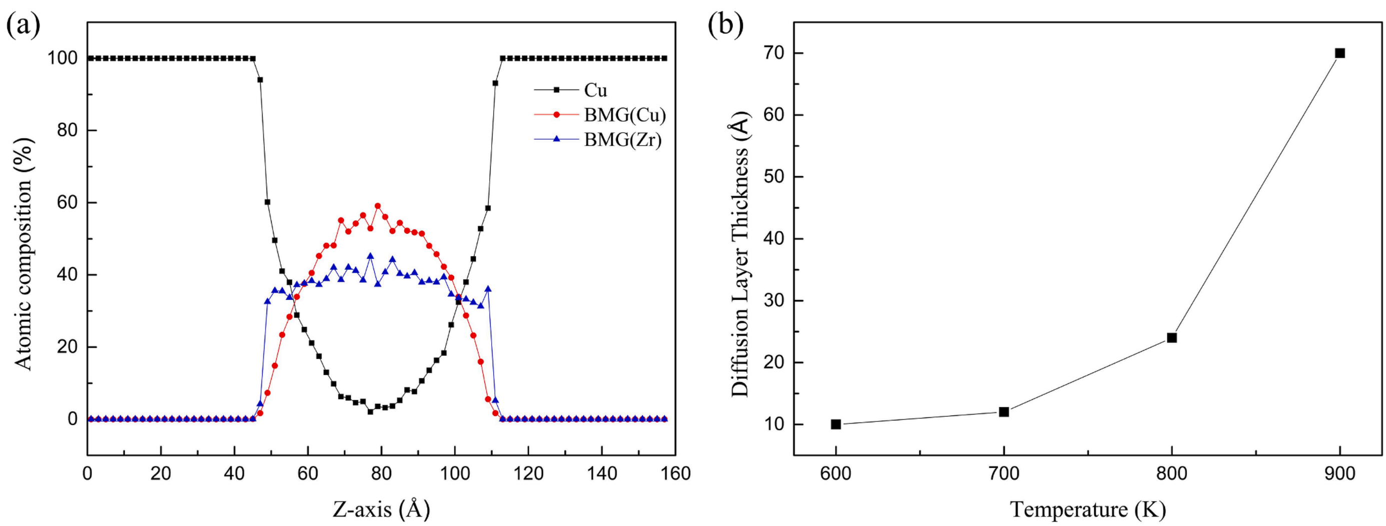

Fig. 10. (a) Concentration distribution of interface atoms at 900 K and (b) Diffusion layer thickness at different temperatures.

addition, in order to further explore the crystallization kinetics of the weld zone and the heat-affected zone, the nucleation rate and growth rate of BMG during the solidification process were calculated. The results showed that the maximum cooling rate could limit the transformation of supercooled liquid to crystal, which made a deeper explanation of the crystallization phenomenon in the process of atomic diffusion. Chen et al. studied the characteristics of dissimilar joints of dendritic reinforced titanium matrix metal glass composite (MGC) and titanium matrix metal alloy glass (MG) under pulsed laser welding [34], and found crystals in the fusion zone β-Ti grains grow from spherical to dendritic, which also reveals the phenomenon of crystal growth caused by cooling rate during laser welding.

5. Conclusions

In this work, Zr-based amorphous alloys and metallic copper were connected by pulsed laser butt welding, and the phase composition, microstructure, chemical composition, phase distribution, and appearance of Zr-based amorphous alloy/copper butt-welded joints were investigated. The following conclusions can be drawn:

(1) Zr_2Cu, Zr_2Ni and other crystalline phases appear on the welding interface between zirconium based amorphous alloy and copper. A large amount of Cu elements diffused into the Zr-based BMG, forming a Cu-rich region in the Zr-based BMG. A small amount of Zr, Ni, and Ti diffused to the Cu side, forming oxides such as ZrO_2, TiO_2, and CuO on the surface of the sample, these high levels of oxides cause the volt potential at the interface to be higher than that of the parent metal on both sides.
(2) A very fine Cu grain structure is displayed in the amorphous alloy near the amorphous alloy/Cu interface, while other regions still show amorphous state and grain growth occurs on the Cu side near the interface.
(3) The microhardness of the welded joints shows high hardness values at the amorphous alloy and lower hardness values on the Cu side near the interface. This is because the diffusion of Cu into the amorphous alloy plays the role of dispersion strengthening, resulting in the hardness of the diffusion layer being higher than that of the base metal of the amorphous alloy, and the nano-hardness of the diffusion layer is also higher than that of the base metal on both sides.

CRediT authorship contribution statement

Wenzhe Zhang: Software, Visualization, Data Curation, Writing – original draft; **Jiankang Huang:** Conceptualization, Funding acquisition, Resources, Supervision, Writing – review & editing; **Yanqin Huang:** Methodology, Investigation, Experiment; **Xiaoquan Yu:** Investigation, Translate, Language revision; **Ding Fan:** Conceptualization, Methodology, Resources, Supervision.

Disclosure statement

No potential conflict of interest was reported by the authors.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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